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ABSTRACT

The condition of road surfaces is a significant factor in the safety of transportation and the performance of vehicles. The traditional road inspection methods are mostly manual, time- consuming, and expensive; they are also unsuitable for continuous monitoring in real-world driving conditions. This project proposes a real-time road anomaly detection system utilizing computer vision and deep learning techniques on an edge computing platform. Road anomalies such as potholes and unexpected obstacles are automatically detected using the Raspberry Pi 4 B+ based on Quad core Cortex-A72 (ARM v8) and image is processed using dashcam video footage. The object detection models were trained on a public dataset from Roboflow, with the addition of necessary preprocessing and data augmentation to enhance performance under different lighting and environmental conditions. Among the lightweight architectures evaluated, YOLOv5n was selected. During model training, GPU acceleration was used, and the trained network was converted into an edge-optimized format either with ONNX Runtime or TensorFlow Lite for deployment. The system integrates the optimized model with an OpenCV-based video processing, supporting near real-time inference at frame rates greater than or equal to 5 FPS. The standard metrics such as Precision, Recall, F1-Score, and mean Average Precision (mAP) were used for performance evaluation, which indicates a reliable anomaly detection with a small number of false positives.Thus providing a scalable and practical solution for real-time road condition assessment and smart transportation applications.

INTRODUCTION

Road surface conditions play a vital role in ensuring safe and comfortable transportation. Defects such as potholes, cracks, and unexpected obstacles can lead to accidents, vehicle damage, and traffic disruptions. Traditional road inspection methods mainly depend on manual surveys or specialized monitoring vehicles, which are time-consuming, costly, and limited in coverage. These limitations highlight the need for automated systems capable of continuously monitoring road conditions in real time. With the rapid advancement of computer vision and deep learning technologies, automated detection of road anomalies has become feasible. These technologies enable efficient analysis of visual data captured from moving vehicles, supporting timely identification of road defects and assisting authorities in maintenance planning and safety improvement.

This project presents a real-time road anomaly detection system implemented on an edge computing platform using Raspberry Pi 4 Model B+. The system captures road video through a dashcam and processes it using a lightweight YOLOv5n object detection model trained on a Roboflow dataset with preprocessing and augmentation techniques. The trained model was optimized for edge deployment and integrated with OpenCV to achieve near real-time inference performance. Standard evaluation metrics including Precision, Recall, F1-score, and mean Average Precision were used to assess model performance, demonstrating reliable anomaly detection with low false positives. The proposed system provides a cost-effective and scalable solution for continuous road monitoring and supports the development of intelligent transportation systems.

LITERATURE SURVEY

Recent studies in road surface monitoring have increasingly focused on automated detection techniques using computer vision and deep learning to overcome the limitations of traditional manual inspection methods. Early approaches relied on image processing techniques such as edge detection, texture analysis, and threshold-based segmentation for identifying potholes and cracks, but these methods were highly sensitive to lighting variations and complex road environments. With the advancement of deep learning, convolutional neural network (CNN)-based models have been widely adopted for road anomaly detection due to their ability to learn robust feature representations. Several researchers have utilized object detection frameworks such as Faster R-CNN, SSD, and YOLO for real-time pothole and obstacle detection, achieving improved accuracy and detection speed. Lightweight variants of YOLO have been particularly explored for deployment on embedded and edge devices, enabling on-board processing without relying on cloud infrastructure. Additionally, the integration of edge computing platforms like Raspberry Pi with optimized deep learning models and OpenCV-based video processing has demonstrated promising results for continuous road monitoring applications. These developments indicate that combining lightweight deep learning models with edge computing provides an effective and scalable solution for real-time road anomaly detection in intelligent transportation systems.

SYSTEM ARCHITECTURE

The proposed road anomaly detection system follows a modular architecture consisting of the following components:

- **Video Input Module (Dashcam Camera)**
- **Frame Extraction & Preprocessing Module (OpenCV)**
- **Anomaly Detection Module (YOLOv5n Model)**
- **Edge Processing Module (Raspberry Pi 4 Model B+)**
- **Post-processing & Visualization Module**
- **Storage/Alert Module**

The workflow is as follows:

1. The dashcam mounted on the vehicle continuously captures road surface video during driving.
2. The captured video stream is fed to the Raspberry Pi, where OpenCV extracts individual frames and performs basic preprocessing such as resizing and normalization.
3. The preprocessed frames are passed to the optimized YOLOv5n model for real-time detection of road anomalies such as potholes and obstacles.
4. The Raspberry Pi performs inference locally as an edge device, enabling offline and low-latency processing without cloud dependency.
5. Detected anomalies are post-processed to draw bounding boxes and labels on frames for visualization.
6. The processed output can be displayed, stored, or used to trigger alerts for further analysis.
   This architecture enables efficient real-time road monitoring with low computational overhead, making it suitable for practical deployment in intelligent transportation environments.

HARDWARE SETUP

The hardware components used in this project include:

• Raspberry Pi 4 Model B+ (4GB RAM) for edge processing and real-time inference
• Dashcam / USB Camera for continuous road video capture
• MicroSD Card for operating system, model storage, and data logging
• Display Monitor (optional) for visualizing detection results
• Power Supply (5V, 3A adapter) to power the Raspberry Pi
• Mounting setup for camera to ensure stable video acquisition during driving
• Connecting cables and accessories for system integration

These hardware components collectively support video acquisition, edge computation, visualization, and reliable operation of the proposed real-time road anomaly detection system.

SOFTWARE IMPLEMENTATION

The software stack includes:

• Raspberry Pi OS as the operating system for edge device operation
• Python programming language for system development and integration
• OpenCV library for video capture, frame extraction, and preprocessing
• YOLOv5n deep learning model for road anomaly detection
• Ultralytics framework for model training and inference support
• ONNX Runtime / TensorFlow Lite for optimized edge deployment
• NumPy library for numerical operations and data handling

Implementation Steps

1. Road video is captured continuously using the camera through OpenCV.
2. Video frames are extracted and preprocessed by resizing and normalization.
3. The preprocessed frames are passed to the trained YOLOv5n model for anomaly detection.
4. The model performs inference on the Raspberry Pi to identify potholes and obstacles.
5. Detected anomalies are post-processed to generate bounding boxes and class labels.
6. The annotated frames are displayed or stored for monitoring and analysis.
7. The system operates continuously to provide real-time road condition assessment during vehicle movement.

RESULTS

The developed system was able to successfully implement real-time road anomaly detection on Raspberry Pi hardware. The system accurately detected road defects such as potholes and obstacles from dashcam video footage using the optimized YOLOv5n model. Edge deployment enabled continuous monitoring without reliance on cloud connectivity, making the system suitable for practical on-road usage. The integration of OpenCV-based video processing with the deep learning model provided stable near real-time inference performance during vehicle movement.

Observations

• Average inference speed: ≥ 5 FPS on Raspberry Pi 4 Model B+
• Detection accuracy: High precision with minimal false positives
• Model loading time: Few seconds during system startup
• Performance under varying lighting: Reliable detection with preprocessing and augmentation support
• Edge processing capability: Stable operation without internet dependency

The system demonstrated consistent real-time performance and effective detection capability, validating its applicability for smart transportation and automated road condition monitoring.

OPTIMIZATION TECHNIQUES

Several optimizations were implemented to improve system performance:

Model Optimization

• Selection of lightweight **YOLOv5n architecture** suitable for edge deployment
• Model conversion to **ONNX Runtime / TensorFlow Lite** for faster inference
• Input image resizing to reduce computational load

Video Processing Optimization

• Frame skipping strategy to balance detection speed and accuracy
• Efficient frame capture using OpenCV streaming pipeline
• Resolution adjustment to maintain real-time performance

Embedded Optimization

• Utilization of Raspberry Pi GPU/CPU resources for accelerated inference
• Preloading of detection model to avoid repeated initialization delay
• Local edge processing to eliminate network latency

System Optimization

• Data augmentation during training to improve robustness under lighting variations
• Confidence threshold tuning to reduce false detections
• Efficient memory management for stable continuous operation

These optimizations improved overall system responsiveness and enabled near real-time anomaly detection at approximately 5 FPS on the Raspberry Pi platform.

CONCLUSION

This project has successfully implemented the design of a real-time road anomaly detection system using computer vision and deep learning on a Raspberry Pi edge computing platform. The proposed system was able to effectively detect road surface anomalies such as potholes and obstacles from dashcam video footage without relying on cloud-based processing. The integration of a lightweight YOLOv5n model with OpenCV-based video processing enabled near real-time inference while maintaining acceptable detection accuracy on resource-constrained hardware. The system demonstrated reliable performance under varying environmental conditions, highlighting the feasibility of deploying deep learning–based road monitoring solutions on embedded platforms for intelligent transportation applications.

The developed system provides a cost-effective and scalable approach for continuous road condition assessment, which can assist in improving road safety and maintenance planning. The project also validates the potential of edge computing in reducing latency and ensuring uninterrupted operation during field deployment. Future enhancements of this work may include expanding the dataset to cover diverse road scenarios, improving detection accuracy through advanced model optimization techniques, integrating GPS for geo-tagging detected anomalies, and enabling wireless communication for centralized monitoring. These improvements can further enhance the practicality and impact of the proposed system in smart city and transportation infrastructure management.